



INSTITUT TEKNOLOGI DEL

MATERI PRAKTIKUM

4233201 – Kecerdasan Buatan

Semester Ganjil Tahun Ajaran 2026/2027

Session Date	: 22 September 2026
Week/Session	: 3/3
Topic	: Polynomial Regression
Practicum Goals	: Mahasiswa mampu melatih model regresi, menentukan persamaan garis regresi dan mengevaluasi performa model
Duration	: 2 x 50 minutes
Place of Submission	: ecourse
Submission Deadline	: Akhir Sesi

```
In [1]: import pandas as pd
path = "Ice_cream selling data.csv"
df = pd.read_csv(path)
print(df.head())
```

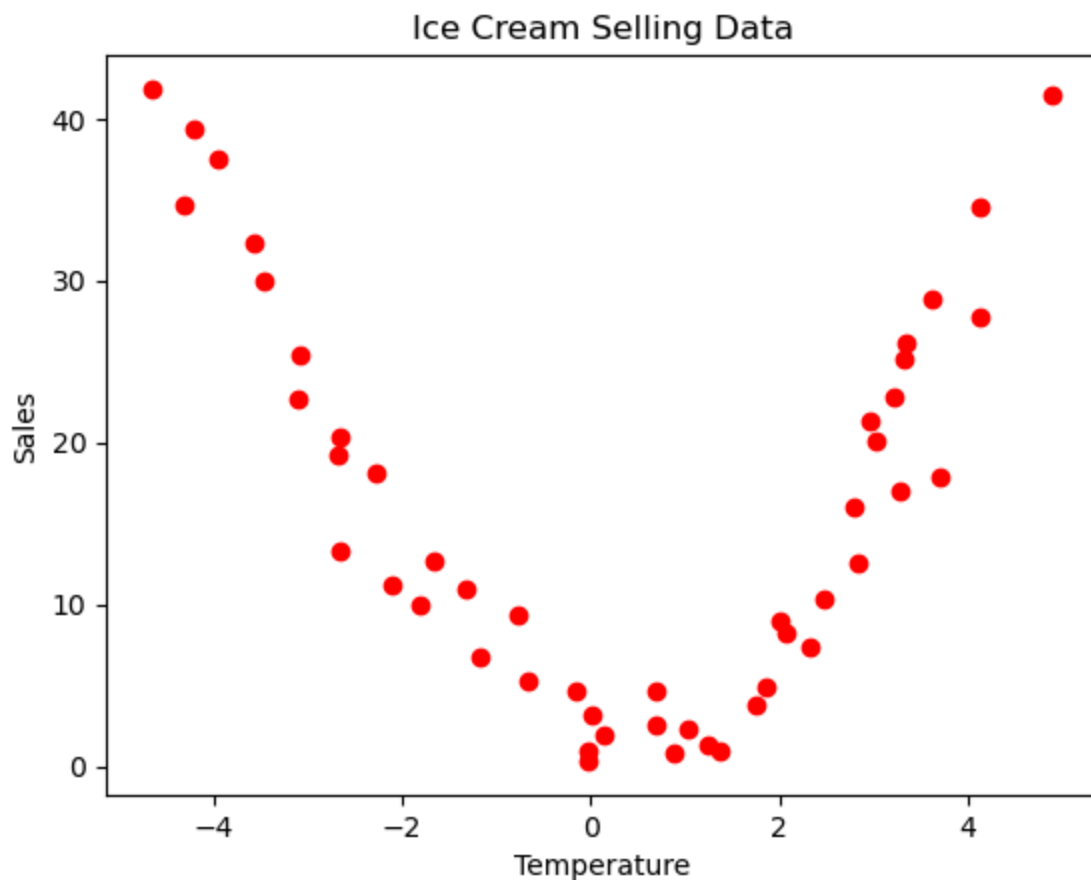
	Temperature (°C)	Ice Cream Sales (units)
0	-4.662263	41.842986
1	-4.316559	34.661120
2	-4.213985	39.383001
3	-3.949661	37.539845
4	-3.578554	32.284531

```
In [2]: X = df.iloc[:, :-1].values
y = df.iloc[:, -1].values
```

```
In [3]: import matplotlib.pyplot as plt

plt.scatter(X, y, color = 'red')
plt.title('Ice Cream Selling Data')
plt.xlabel('Temperature')
plt.ylabel('Sales')
plt.show
```

```
Out[3]: <function matplotlib.pyplot.show(close=None, block=None)>
```

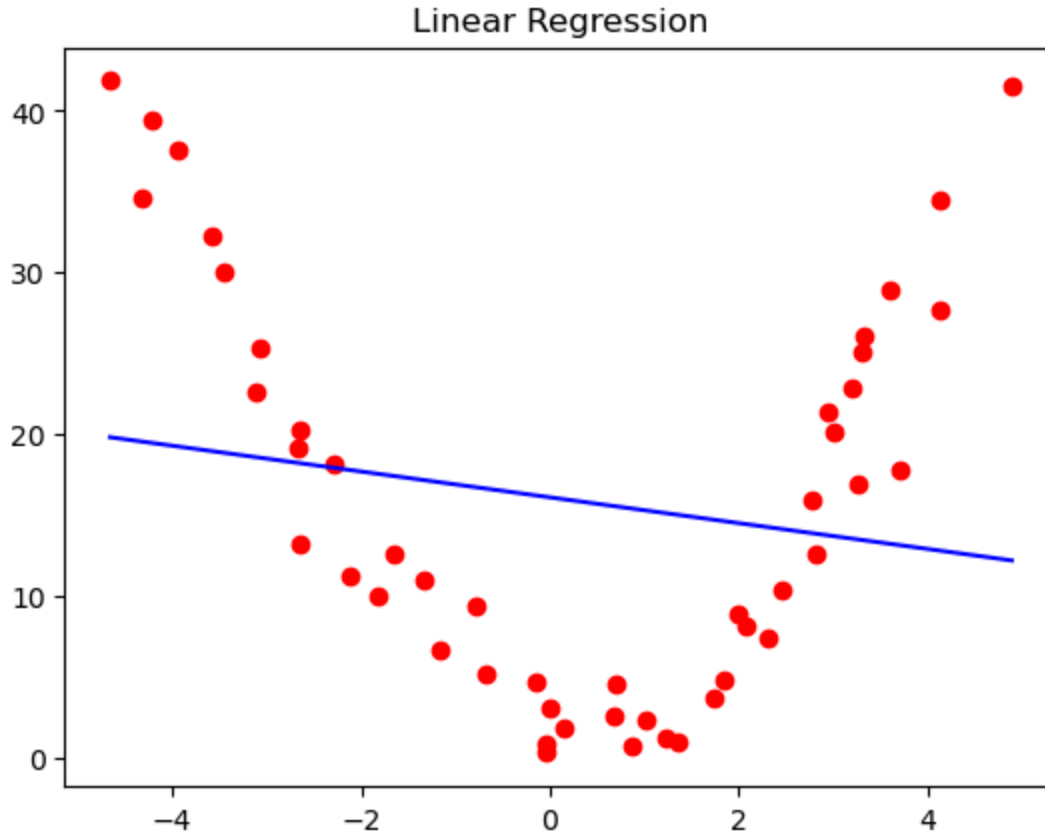


```
In [4]: from sklearn.linear_model import LinearRegression
```

```
linear_model = LinearRegression()
linear_model.fit(X,y)

plt.scatter(X, y, color = 'red')
plt.plot(X, linear_model.predict(X), color = 'blue')
plt.title('Linear Regression')
```

Out[4]: Text(0.5, 1.0, 'Linear Regression')



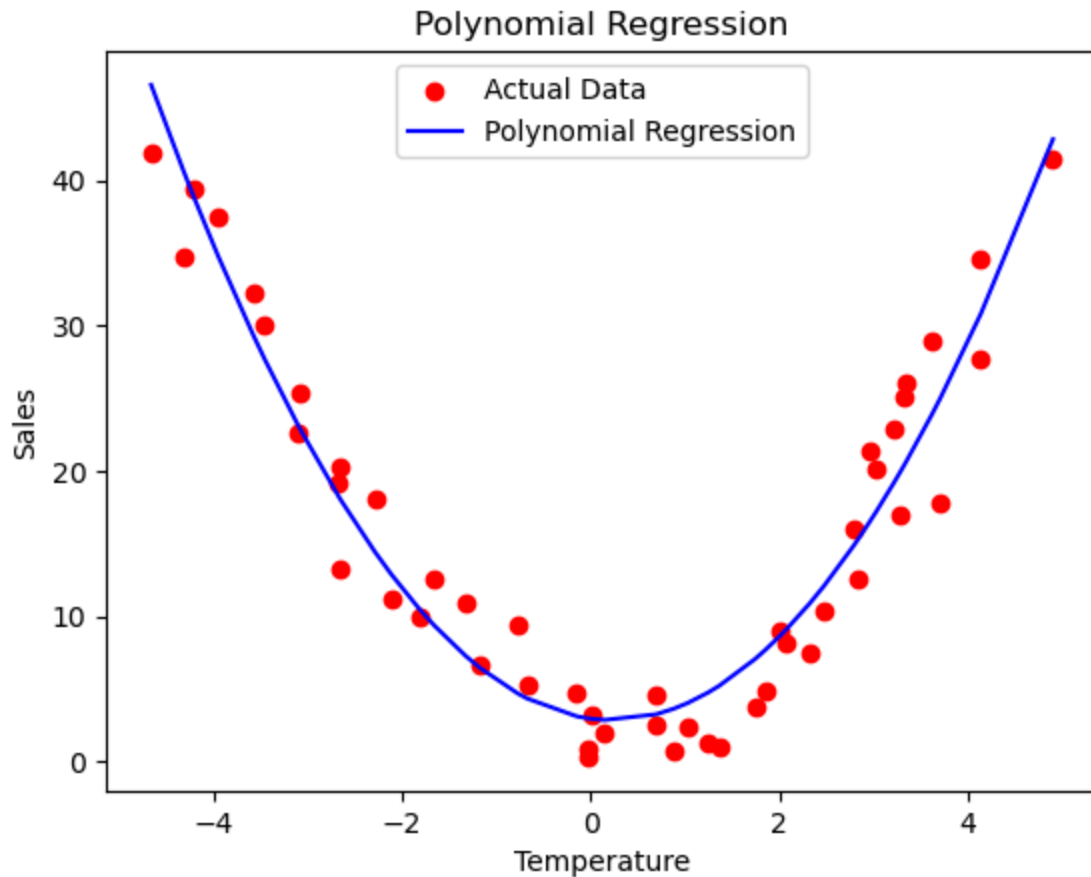
In [5]: `from sklearn.preprocessing import PolynomialFeatures`

```
# Buat Polynomial Features
poly_reg = PolynomialFeatures(degree=2)
X_poly = poly_reg.fit_transform(X)

# Buat model Linear Regression
linear_model2 = LinearRegression()
linear_model2.fit(X_poly, y) # Latih model dengan data polynomial

# Gunakan predict() untuk mendapatkan y_pred_poly
y_pred_poly = linear_model2.predict(X_poly)

# Plot Polynomial Regression
plt.scatter(X, y, color='red', label="Actual Data")
plt.plot(X, y_pred_poly, color='blue', label="Polynomial Regression")
plt.title('Polynomial Regression')
plt.xlabel('Temperature')
plt.ylabel('Sales')
plt.legend()
plt.show()
```



```
In [6]: print('Intercept: ', linear_model2.intercept_)
print('Coefficient: ', linear_model2.coef_)

print('Equation: y = ', linear_model2.intercept_, '+', linear_model2.coef_[1], 'x +
```

```
Intercept: 2.9517741579934498
Coefficient: [ 0.          -0.82468167  1.82952623]
Equation: y = 2.9517741579934498 + -0.8246816702106482 x + 1.829526229670208 x²
```

Polynomial Regression dengan lebih dari satu predictor

```
In [7]: dataset = pd.read_csv('manufacturing.csv')
print(dataset.head())
```

	Temperature (°C)	Pressure (kPa)	Temperature x Pressure \
0	209.762701	8.050855	1688.769167
1	243.037873	15.812068	3842.931469
2	220.552675	7.843130	1729.823314
3	208.976637	23.786089	4970.736918
4	184.730960	15.797812	2918.345014

	Material Fusion Metric	Material Transformation Metric	Quality Rating
0	44522.217074	9.229576e+06	99.999971
1	63020.764997	1.435537e+07	99.985703
2	49125.950249	1.072839e+07	99.999758
3	57128.881547	9.125702e+06	99.999975
4	38068.201283	6.303792e+06	100.000000

```
In [8]: X = dataset.iloc[:, 2:5].values
y = dataset.iloc[:, 5].values
X
```

```
Out[8]: array([[1.68876917e+03, 4.45222171e+04, 9.22957596e+06],
 [3.84293147e+03, 6.30207650e+04, 1.43553672e+07],
 [1.72982331e+03, 4.91259502e+04, 1.07283887e+07],
 ...,
 [3.95630467e+03, 6.26576910e+04, 1.40595745e+07],
 [4.97723476e+03, 5.71959855e+04, 9.13403620e+06],
 [3.97489712e+03, 4.10923929e+04, 4.35891253e+06]])
```

```
In [9]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

```
In [10]: linear_model = LinearRegression()
linear_model.fit(X_train, y_train)

poly_reg = PolynomialFeatures(degree=2)
X_train_poly = poly_reg.fit_transform(X_train)
X_test_poly = poly_reg.transform(X_test)

linear_model2 = LinearRegression()
linear_model2.fit(X_train_poly, y_train)

y_pred = linear_model2.predict(X_test_poly)
```

```
In [11]: from sklearn.metrics import r2_score
score=r2_score(y_test,y_pred)
score
```

```
Out[11]: 0.8199500002058089
```

Tugas

Analisis perbedaan R-square score polynomial derajat 1, 3, 5, untuk Polynomial Regression dengan lebih dari satu predictor